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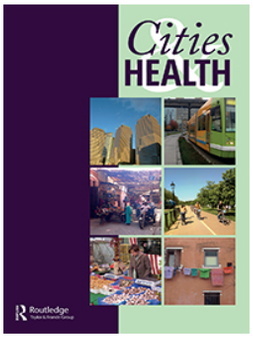


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Acoustic quality and health in urban environments (SALVE) – a pilot study in the metropolitan Ruhr region, Germany

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ABSTRACT

Acoustic quality has long been a concern in the disciplines of architecture, building technology, and construction engineering. In health research, acoustic quality has been understood as a health risk factor of noise and, thus, limited to reducing sound decibel levels. Noise mitigation measures exclusively aim to reduce noise levels below a threshold. In contrast, acoustic environments are a composition of all acoustic events of the natural and physical setting, typically determined by sound level, frequency, time, and space. Thus, soundscape ecology provides a suitable methodical approach to analyze associations between acoustic quality, the built environment, and human health. This paper debates how SALVE, a two-year interdisciplinary pilot study, may fill the knowledge gap and contribute to sound and health research. SALVE aims to identify criteria for human health-promoting acoustic environments. By making annual direct and automated auditory measurements of a robust land-use sample, a large multi-seasonal urban sound dataset will be established. This data is merged with health data from the population-based Heinz Nixdorf Recall study and will form a starting point for an innovative approach to better understand the impacts of acoustic quality on urban public health beyond noise protection.

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Acoustic environment; soundscape; sound index; acoustic quality; pilot study; health

Introduction

Noise is considered as an environmental burden and a globally recognized health risk (WHO Regional Office for Europe 2011, 2018). Current research is concerned with the possible impacts of noise on health, especially in cities, where a large proportion of the population is exposed (WHO Regional Office for Europe 2011). In particular, health effects of road traffic noise have been well studied (Babisch *et al.* 2005, Barregard *et al.* 2009, Selander *et al.* 2009, Fuks *et al.* 2011, Sorensen *et al.* 2011, 2013, Chang *et al.* 2013, Kalsch *et al.* 2014, Orban *et al.* 2016).

Noise is considered as a loud, unpleasant and rather unwanted sound (Brown 2006, Jarosinska *et al.* 2018, Oxford English Dictionary 2018), which is disruptive and causes disturbances (Guski *et al.* 1999, Oxford English Dictionary 2018).

In general, sound is described as audible vibrations (pressure waves) which can reach a person's ear (Passchier-Vermeer and Passchier 2000, Oxford English Dictionary 2018). Sound pressure levels [decibels (dB)] are used to express sound pressure amplitude, measured in pascal (Pa). In environmental noise research, A-weighted dB (dBA) is a common measure to identify whether certain levels are below recommended noise thresholds that are associated with adverse health effects. A-weighting and C-weighting audible frequencies are used to reflect the response and sensitivity of the human ear (WHO Regional Office for Europe 2018); C-weighting dependent frequency corrections are typically used for peak measurements (WHO Regional Office for Europe 2018). Noise exposure is distinguished between voluntary and involuntary noise exposure. Effects of involuntary noise exposure are associated with feelings of anger, annoyance or disturbance (Guski 1997, Guski *et al.* 1999) and are summarized as noise annoyance. The underlying concept is based on the assumption that noise only becomes noise through subjective perception. Accordingly, both – measured noise and noise perception (noise annoyance) – are important exposure parameters in noise research (Guski 1999, Guski *et al.* 1999).

Raimbault and Dubois (2005) emphasize that the sound pressure level alone is not sufficient to determine the quality of sound, but that additional factors such as perception, meaning and mixtures of sounds are relevant and should be considered (Raimbault and Dubois 2005). Research dealing with noise and health effects recommends strategies for noise reduction and human protection from noise. Per se noise management is necessary (Schulte-Fortkamp 2012), however, this approach is not sufficient to create health-promoting environments (Raimbault and Dubois 2005, Kang *et al.* 2016). Less noise does not generally mean more

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pleasant sound resulting in better overall health and quality of life (Yang and Kang 2005a, 2005b, Brown 2006, Andringa and Lanser 2013, Van Kempen *et al.* 2014). Sound that is very unwanted and annoying for some people can be more pleasant and wanted for others (Bronzaft 2016). Noise mitigation is essential in order to achieve a high-quality acoustic environment. Nevertheless, until today it is not completely clear on what is required to achieve and preserve high-quality acoustic environments. For example, a high-quality acoustic environment could be achieved by promoting biophonic sounds (with the co-benefit of biodiversity increase) and by limiting anthropogenic noise.

The concept of soundscape ecology which encompasses the study of spatial-temporal heterogeneity of sounds in different landscapes (Pijanowski *et al.* 2011, Farina *et al.* 2014, Farina 2014a) provides a suitable methodological approach to analyze the relationships between sonotopes (patterns), the built environment and thus human health. Farina (2014b) defines sonotopes as the spatial overlap and mixture of different sounds. Pijanowski *et al.* (2011, p. 204) define soundscapes as 'all sounds, those of biophony, geophony, and anthrophony, emanating from a given landscape to create unique acoustical patterns across a variety of spatial and temporal scales' (acoustic environment). The same authors recognize within their conceptual framework that interactions between humans and the acoustic environment are essential in both ways: humans can influence the acoustic environment and the acoustic environment can influence human well-being (Pijanowski *et al.* 2011).

In 2014, the term soundscape was officially defined by the ISO 12913–1 as 'an acoustic environment as perceived or experienced and/or understood by a person or people, in context', thus identifying soundscapes as a 'perceptual construct' (International Organization for Standardization 2014). Until today, different definitions exist in regard to 'soundscapes' depending on the discipline and field of research (Pijanowski *et al.* 2011). Overall, soundscape research goes beyond the concept of noise and is concerned with all sorts of physical sound within an environment and how this sound is perceived by humans (Kang *et al.* 2016, Schulte-Fortkamp 2018).

For our purpose, we will use the term acoustic environment when considering and measuring sounds in the environment. The term soundscape is used when addressing the perception of such acoustic environments.

From a perceptual perspective (psychoacoustics), soundscape research has become more important when considering negative and positive effects on human health (Aletta *et al.* 2018). A systematic review of the literature from 1991 to 2018 investigated the associations between positive health-related effects and the perceptual soundscape construct (Aletta *et al.* 2018). Based on a qualitative synthesis approach, the authors concluded that experiencing positive soundscapes is associated with positive health outcomes (Aletta *et al.* 2018). It is currently being discussed that certain environments facilitate restoration due to their high acoustic quality (Irvine *et al.* 2009, Lercher *et al.* 2016). For example, studies show that greener environments are associated with stress reduction (Ulrich *et al.* 1991), and better self-rated health (Orban *et al.* 2017), vice versa. Negative soundscape perception has been shown to be associated with decreased mental health and severe annoyance (Lercher *et al.* 2016).

By analyzing soundscapes in an urban area in Rostock, Germany Liu *et al.* (2013) came to the conclusion that landscape characteristics play an important role in the composition and perception of soundscapes (Liu *et al.* 2013). They concluded that the concept of acoustic quality should take place in urban planning at an early stage by considering the importance of natural sounds for increased quality of life and thus environmental health (Liu *et al.* 2013). This goes in line with early work from Southworth (Southworth 1969) and Schafer (Schafer 1977) who addressed the built environment and its sound. Furthermore, they emphasized the way cities need to be built in order to guarantee a high-quality acoustic environment (Kang *et al.* 2016). This also shows that the broader context of environmental quality should be considered when looking at the way soundscapes could potentially link with health (Van Kempen *et al.* 2014).

Assuming that pleasant and wanted sound within the urban environment has a beneficial impact on health and well-being of the population, it can be decisive for city planning and health research to further include the concept of soundscapes (Kang *et al.* 2016). Due to urbanization, especially in large cities, the positive effects of soundscapes can be of great interest, as environments of high acoustic quality are needed (Raimbault and Dubois 2005, Van Kempen *et al.* 2014). So far, knowledge about indicators that determine good urban acoustic quality as well as knowledge about the relationship between high urban acoustic quality and human health is missing (Van Kamp *et al.* 2016). It is important to stress that the concept of high acoustic quality is different from that of quietness (Brown 2006). High-quality acoustic outdoor environments are not necessarily silent or quiet but are often considered as more pleasant and peaceful. Therefore, it is important to further focus on the complex issue of wanted sounds compared to unwanted sounds and what is considered as high acoustic quality in urban environments (Brown 2006). Especially, in cities, it is necessary to identify environments of high acoustic quality in order to preserve the diversity of people' experiences and thus the quality of life (Brown 2006). Brown (2006, p. 8)

stresses that empirical evidence is needed in order to identify "the benefits that people derive from the experience of areas of high acoustic quality".

In order to generate further knowledge, the interdisciplinary two-year pilot study 'acouStic quAlity and heaLth in urban enVironmEnts – analysis of the interrelationships between soundscapes and health (SALVE)' was initiated in the Metropolitan Ruhr region in Germany, involving disciplines from public health and spatial planning. This paper aims to debate how SALVE can fill the knowledge gap and contribute to the field of acoustic environment, sound and health research. SALVE serves as a starting point for an innovative approach to understand the impacts of acoustic quality on urban public health looking beyond noise protection.

The novelty of SALVE derives from the idea of treating urban land use as a so-called 'human habitat', and attempting to use sound as an indicator of habitat quality, similar to the use of bioacoustic indices as a measure of the quality of non-human habitats. Furthermore, a temporally longitudinal set of recordings will be systematically gathered from the urban environment with both automated and direct recordings, which can be used for a wide array of follow-up studies. Overall, the application of sound indices in urban environments will provide a deeper understanding of sound than traditional decibel focused indices used in noise-related studies.

SALVE project

The two – year pilot study SALVE started in October 2018 in Bochum. By direct and automated auditory measurements, a large multi-seasonal urban sound dataset is to be established in the Metropolitan Ruhr region and then merged with comprehensive health-related data from the longitudinal, population-based Heinz Nixdorf Recall (HNR) study. Prospectively, knowledge will be gained for the development of solution-based health-promoting strategies in spatial planning. Furthermore, residents in Bochum will be asked for their soundscape preferences in order to understand how people experience sound in urban environments (Urban soundscape questionnaire, Figure 1).

Aims of SALVE

The aims of SALVE are to measure spatial-temporal differences in acoustic environments, taking into account the built environment, defined as land use types, and temporal changes. Sound indices as used in Soundscape Ecology will be calculated and assigned to specific land-use types. Spatial-statistical models will be further developed to analyze the health effects of identified types of soundscapes in urban neighborhoods. As part of the pilot study, knowledge on data protection issues, acceptance, and feasibility of measurement devices is to be gathered.

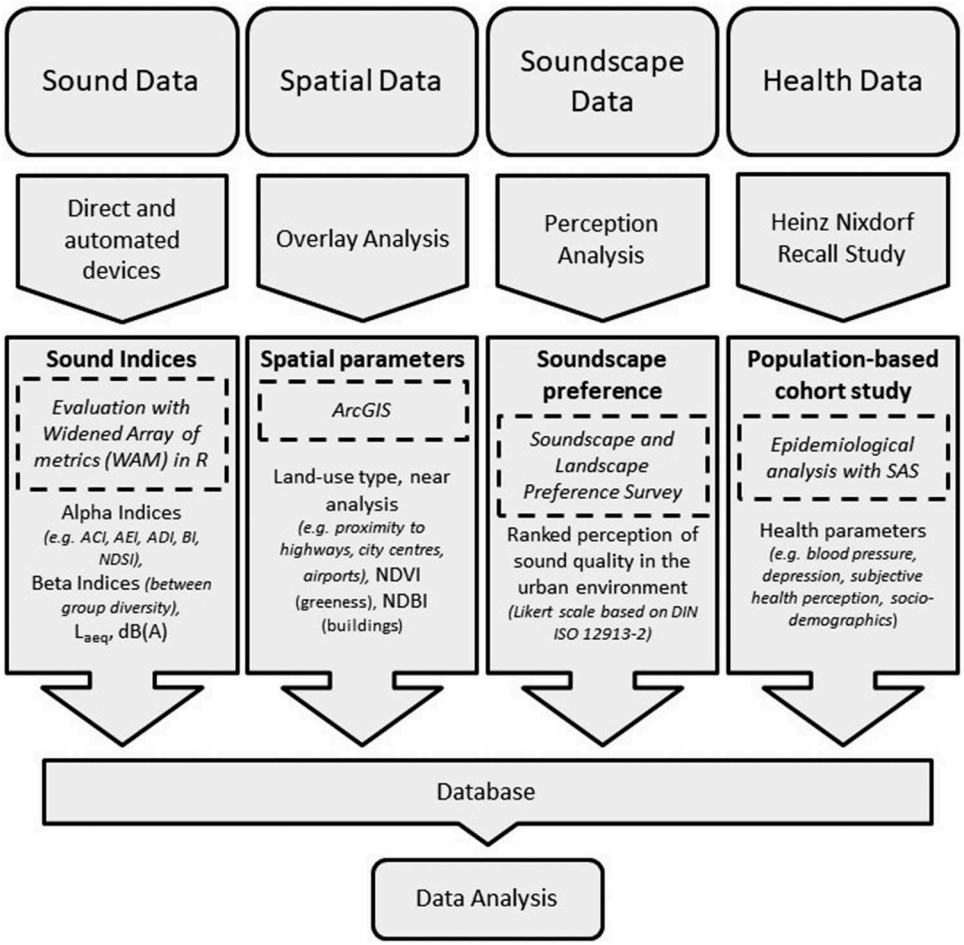


Figure 1. Process of data analysis.

Knowledge on identified soundscapes will be used to identify how people experience and perceive these sounds within their environment.

Study design

Study area

The Metropolitan Ruhr Region is located in the federal state of North Rhine-Westphalia in Germany. It is the largest urban agglomeration in Germany and one of the five largest in Europe, where approximately 5.2 million inhabitants live (The Regional Association Ruhr 2018).

Bochum, which has a size of 145.2 km2, is one of the larger cities centered in the southeast of the Ruhr region. In 2018, about 370,000 inhabitants were living in Bochum, reaching a population density of 2,550/km2 (City of Bochum 2018).

Study population

Health data will be based on the Heinz Nixdorf Recall Study (HNR), a population-based cohort study from three large cities (Bochum, Essen and Mülheim/Ruhr) in the Metropolitan Ruhr Region in the West of Germany. The study has been described elsewhere in detail (Schmermund *et al.* 2002). The study is designed as an individual-based observational study with a sample size of 4814 participants aged 45–74 years (men 49.8%) who were randomly selected from population registries. The data sets include individual clinical, socio-economic and psychosocial as well as contextual parameters. By linking HNR health-related individual data with environmental data, the research could already identify spatial relationships. Living near busy roads is associated with a higher prevalence of coronary heart disease as well as the prevalence of advanced coronary atherosclerosis (Hoffmann *et al.* 2006). Further analyses show that simultaneous exposure to traffic congestion and low social status is associated with high risks for coronary disease (Dragano *et al.* 2009), and that risk factors, such as inflammatory parameters or lack of exercise, are influenced by the urban environment (Dragano *et al.* 2007). High road traffic noise at home is associated with increased depressive symptoms (Orban *et al.* 2016). In contrast, residential greenness within a 100 m radius of people's home is associated with overall neighborhood satisfaction and their overall self-rated health (Orban *et al.* 2017).

Sound and spatial measurements

The study will involve an extensive collection of spatially and temporarily varying sound data in the study area of Bochum. Twenty-four recording devices that are installed in the field and follow an automated aural procedure will record the surrounding sound twice an hour for 3 min each for 365 days. In addition, recording devices that are carried into the field by researchers and follow a direct aural procedure will also be used to record the surrounding sound 4 times a year (each season) for 5 min anytime between 9 am and 6 pm. The direct sound measurement devices will be strategically located so that the acoustic environment is captured in all land use categories in the study area. The location of the automated sound measurements devices follows a similar strategy but uses an equal number of observations for each land-use type, locating three automated devices for each of the seven residential land use categories in Bochum. One automated device is also located in green spaces adjacent to built-up areas, deciduous forest, and mixed forest to obtain comparative soundscape measures for non-urban land use categories. All sampling uses the Sample Design Tool in ArcGIS (Environmental Systems Research Institute Inc., 2018). While the automated devices can record sounds over daily, monthly, seasonal, or annual periods, the direct aural approach can provide a snapshot of sonotopes in a large geographic range of land uses in the study area. This approach provides (i) a temporally detailed but geographically limited baseline of longitudinal sound data over 1 year for selected land-use types and then (ii) a temporally limited but geographically detailed sound dataset to understand how different locations within the study area deviate from the baseline of automated recordings.

A widened array of metrics (WAM), including 29 calculated indices in R (Ihaka and Gentleman 1996), will be applied to analyze sound data. The WAM approach includes α- indices (within-group) such as acoustic complexity index (ACI), acoustic entropy index (AEI), acoustic diversity index (ADI), bioacoustics index (BI) and normalized difference soundscape index (NDSI), β- indices (between-group), and dB measures (Sueur *et al.* 2014, Margaritis *et al.* 2018, Sueur 2018) to define and differentiate observations, delineate and cluster sonotopes geographically and compare human health outcomes with sound distributions. Alpha-indices describe attributes of different sonotopes (heterogeneity, evenness, complexity) and β -indices describe dissimilarity/similarity between sonotopes. Thereby the concept framework of this pilot study lies in landscape ecology with land use as the patch (Forman and Godron 1986), in soundscape ecology with each patch a sonotope within an urban soundtope (Farina 2014) and ecoacoustics with the presence of many species, including humans (Farina and Gage 2017).

Spatial parameters such as proximity of recording points to highways, city centers, airports, availability of surrounding green areas (Normalized Difference Vegetation Index, NDVI) and land use mix will be incorporated. For example, the NDVI is used as a measurement of greenness within a certain area.

Data analyses

Figure 1 depicts the process of data analysis in SALVE. The majority of α- and β- indices will be calculated in R Studio, using the packages sound ecology, seewave, stats, and FactoMineR. These packages in R cover 29 of 36 indices proposed for the WAM. For the remaining indices, especially the α-indices a category of analysis that will explain the composition and complexity of single recordings, Kaleidoscope will be of use. Kaleidoscope is a software tool that employs machine learning to cluster and visualize sounds within a recording based on frequency composition. Sound indices will be interpolated for the city of Bochum with ArcGIS.

The calculated sound indices (exposure) will be examined for associations with health-related variables of the participants (outcome) in the HNR study in Bochum. This data will be included in the statistical models and further stratified analyses will be carried out. Furthermore, recruited as a convenient sample, residents in Bochum rank their perception preferences for different soundscapes on a Likert scale.

Conclusion and outlook

Sound in urban areas has been a concern of architecture, building technology and construction engineering for years. In the context of health research, the sound has been restricted to the health risk factor noise, thus reduced to sound decibel levels. Accordingly, noise mitigation measures aim exclusively at the reduction of the sound level below a certain threshold. In contrast, by focusing on the acoustic environment allows identification of indicators reflecting the quality of different sounds taking into account of the built environment. SALVE will describe the acoustic environment with a diverse set of indices and will test the application of these indices in an urban setting. SALVE will provide a deeper understanding of sound than traditional decibel focused indices used in noise-related studies. It will investigate implications of urban acoustic environments and soundscapes for health. Therefore, the pilot study tries to shift away from investigating the negative effects of noise and focus on the positive impacts of sound for health. Additionally, knowledge will be gained for the development of solution-based health-promoting strategies in spatial planning.

SALVE will rely on a novel mix of methodological approaches used in soundscape ecology by physically measuring the sound in the environment (sound indices from ecoacoustics) and to some extent also integrating the aspects of psychoacoustics by including the human perception of specific soundscapes.

A strategic goal is to lay the foundations for an interdisciplinary and innovative research area 'Healthy Urban Soundscapes' which will focus on indicators, associations and evidence-based solutions that foster a sustainable and health-promoting development in cities. In the longer term, we hope to see acoustic environment research as an integral and active part of public urban health and healthy urban planning fields.

Disclosure statement

No potential conflict of interest was reported by the authors.

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Dietwald Gruehn is a full professor of Landscape Ecology and Landscape Planning at the School of Spatial Planning, TU Dortmund University, Germany (since 2006). 2010 he was awarded adjunct professor at the School of Planning, Construction and Design, Michigan State University, USA. From 2014 to 2017 Dietwald Gruehn was adjunct professor at the Graduate School of National Public Policy at Chungnam National University, Republic of Korea. From 2005 to 2006 he was heading the Environmental Planning

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Susanne Moebus is a biologist and epidemiologist. Since 1999 she has been the head of the complementary medicine project area at the Institute for Mecical Infoamtics, Biometry and Epidemiology (IMIBE). In 2004 she became head of the Cardiovascular Epidemiology & Prevention working group. She is also Head of the Centre for Urban Epidemiology, IMIBE, University of Duisburg-Essen (2012). Until 2019 she was the President of the German Society for Social Medicine and Prevention (DGSMP). She is the project leader (PI) for SALVE.

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